

Annual Precipitation Bulletin 2025: certification of the statewide figure

To: Bulletin editor From: Compilation, state climate office Date: 6 October 2026

Certified figure

The statewide annual precipitation for 2025 is certified at **1,141.1 mm**, the mean of the annual totals of **97 qualifying stations**. Neither figure in circulation is certified: the draft table's 864.2 mm across 428 stations and the reviewer's 1,138.7 mm across 59 stations are both set aside.

What the two figures counted

Draft table, 864.2 mm across 428 stations: not certified. It is the mean of a plain sum of each station's daily values, over every station that reported a daily value in 2025. It ignores completeness, so stations that reported for a few weeks pull the mean down, and it ignores the multi day totals that 252 stations use, so rain that fell during those periods is dropped. The same files reproduce it exactly.

Reviewer, 1,138.7 mm across 59 stations: not certified. It keeps only stations with a daily value on all 365 days and sums those values. That treats every day inside a multi day total as unmeasured, which wrongly disqualifies 39 stations whose years are complete once their usable multi day totals are counted, and it keeps both Blue Hill identifiers, which the standard merges. The same files reproduce it exactly.

How the certified figure was built

The population is every station registered to Massachusetts with a 2025 precipitation record in the 2025 partition of GHCN Daily: 430 stations. Values carrying a quality flag were treated as missing (27 stations carry one). A trace is a measured zero. A multi day total was used only with an unflagged day count and a covered period wholly inside 2025; it then counts every covered day as measured, contributes its full amount, and supersedes any daily value inside its period. A station qualifies only when all 365 days are measured; 332 stations fall short. Blue Hill is registered twice at the same coordinates (USW00014753 and USC00190736); the cooperative record, whose precipitation record begins in 1886, is kept and the other dropped. Keeping both would give 98 stations and 1,141.8 mm. Values are in tenths of a millimetre in the files and are reported here in millimetres.

Wettest and driest qualifying stations

Wettest: **Conway 3.4 Wsw (US1MAFR0041)** at 1,440.8 mm. Driest: **Groveland 0.5 Wsw (US1MAES0004)** at 911.1 mm. The full station table is in station_certification_2025.csv.

Boston Logan

Boston Logan International Airport (USW00014739) measured **943.6 mm** in 2025, every day measured. Its record runs from 1936; 89 of the 90 years satisfy the completeness rule and 2003 is left out (2 days not measured). Among the 89 complete years, 2025 ranks **64th wettest**, between 2020 at 954.4 mm and 1944 at 942.1 mm. The station file in the workspace stops on 6 February 2025, so the 2025 year is completed from the 2025 partition; the two agree on every overlapping day.

Sensitivity of the statewide figure

The qualifying station whose removal would move the statewide figure the most is **Conway 3.4 Wsw (US1MAFR0041)**: without it the mean of the remaining 96 stations would be 1,138.0 mm, a move of -3.1 mm.

Why no other figure survives

The standard fixes the population, the treatment of flags, traces and multi day totals, the completeness test, the one record per site rule and the statistic. Applied to the shipped files there is exactly one qualifying set of 97 stations and one mean. Every other figure relaxes one rule: counting multi day periods as unmeasured, ignoring the totals they carry, skipping the completeness test, or counting a site twice. 1,141.1 mm across 97 stations is the only figure the standard and the files allow.